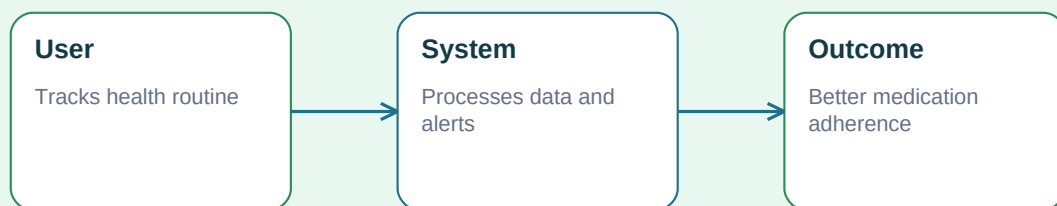


Healthcare Monitoring System

Project Documentation
Stakeholders: User and System

A user-focused web portal for medicine schedules, reminders, inventory, alerts, notifications, profile management, and compliance reporting.



Prepared for project submission and review

1. Introduction

Healthcare Monitoring System is a web application designed to help a user manage daily medicines, reminders, medicine stock, alerts, profile details, notifications, and health compliance reports from a single portal. The system focuses on making routine health tracking simple, timely, and organized.

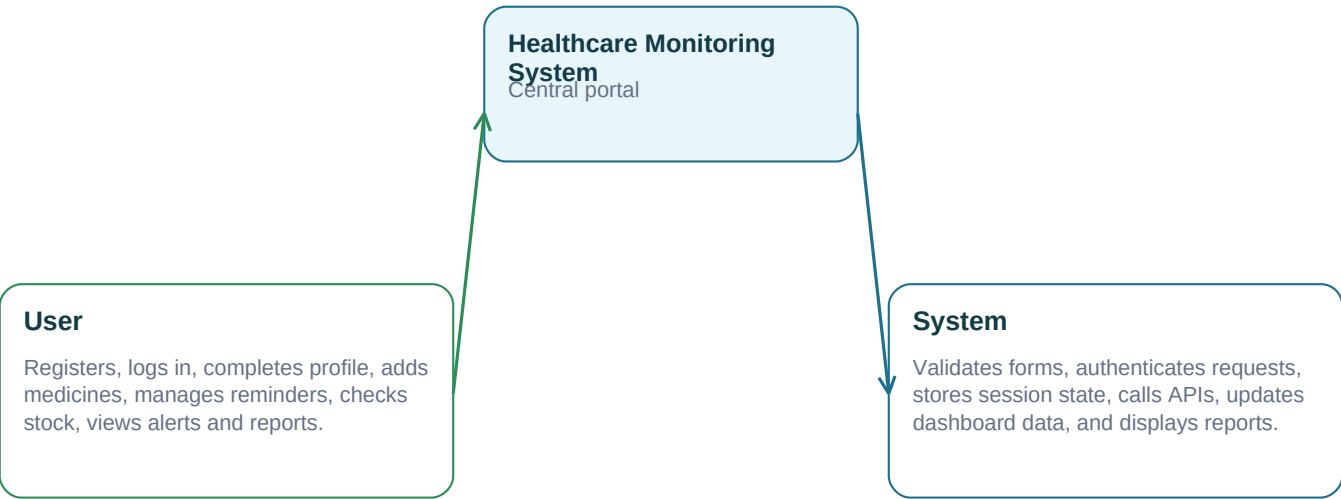
Purpose

This documentation explains the project from the User and System perspective only. It is suitable for project submission, review, and future maintenance.

Scope

- User registration, email verification, login, forgot password, reset password, and logout.
- Profile completion and health detail management.
- Medicine scheduling with dosage, frequency, timing, start date, and notes.
- Reminder tracking with taken, snooze, history, and delete actions.
- Inventory tracking with low-stock and out-of-stock status.
- Alerts, notifications, and compliance reports.

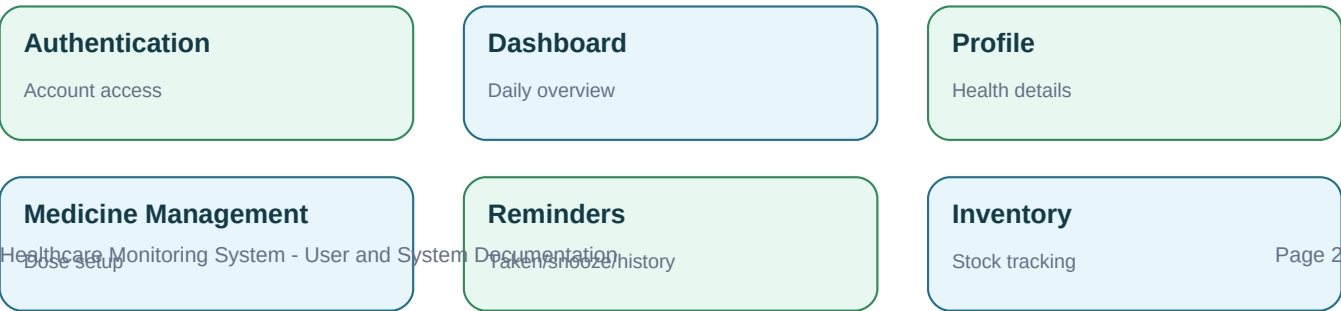
2. Stakeholder Diagram



3. Project Objectives

- Provide a simple responsive portal for personal healthcare and medicine tracking.
- Reduce missed doses through reminder visibility and alerts.
- Help the user maintain medicine availability through stock status.
- Show health compliance through taken and missed dose reports.
- Keep protected requests secured through token-based API access.

4. User Module Map



5. User Journey Flow



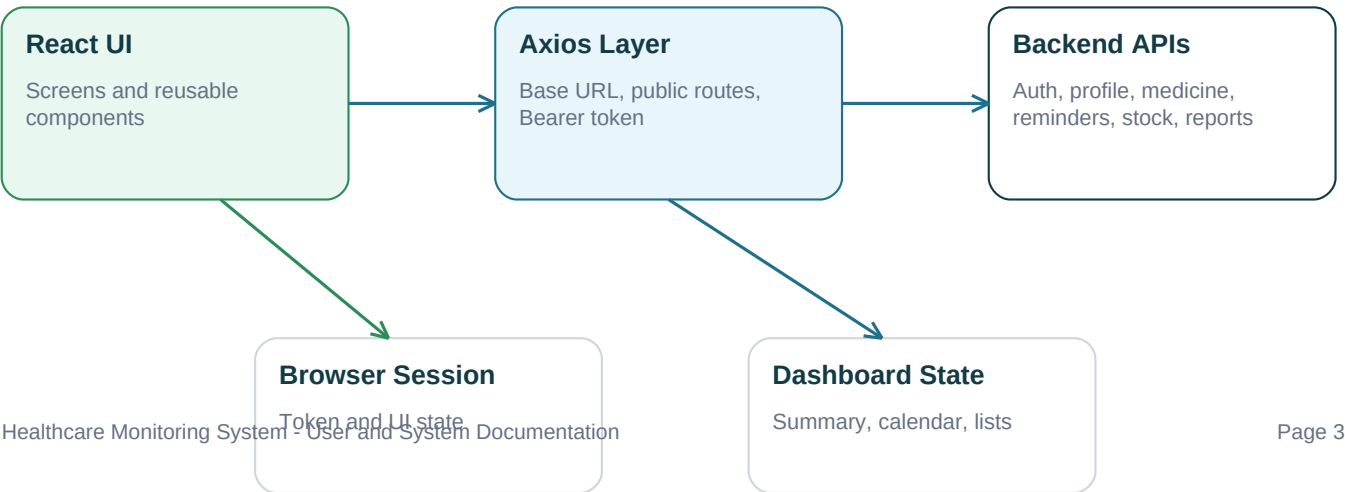
6. Functional Requirements

ID	Requirement
FR-01	User can register with full name, email, mobile number, password, and confirm password.
FR-02	System validates email, mobile number, password strength, and password match.
FR-03	User can login and the system stores authenticated session details.
FR-04	User can complete and update profile details.
FR-05	User can add, edit, search, and delete medicines.
FR-06	System displays today's schedule, reminders, and calendar status.
FR-07	User can mark reminders as taken, snooze reminders, or delete them.
FR-08	User can add, edit, delete, search, sort, and paginate inventory records.
FR-09	User can read alerts and manage notifications.
FR-10	System displays compliance and inventory report statistics.

7. Technology Stack

- React 19 is used for component-based frontend development.
- Vite is used for development server and production build.
- Axios handles API communication and authorization headers.
- Lucide React and React Icons provide UI icon support.
- CSS files define responsive screens, forms, cards, tables, modals, and charts.
- localStorage is used only for browser session state and user interface cache.

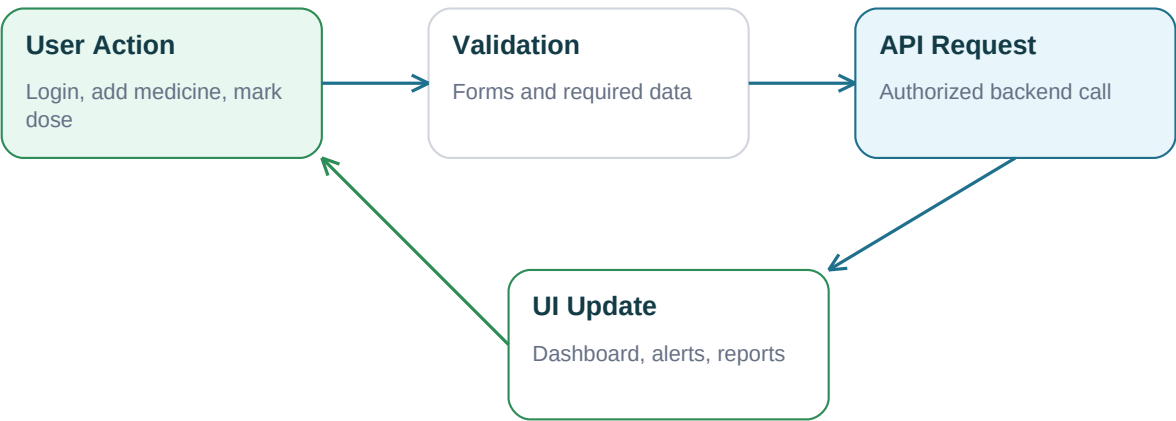
8. System Architecture Diagram



9. API Endpoints Used By User Features

Method	Endpoint	Purpose
POST	/api/auth/register	Create user account.
GET	/api/auth/verify-email	Verify registered email.
POST	/api/auth/login	Authenticate user and return token.
POST	/api/auth/forgot-password	Send reset password link.
POST	/api/auth/reset-password	Reset password using token.
PUT	/api/profile	Complete or update user profile.
GET/POST/PUT/DELETE	/api/medicines	Manage medicine records.
GET/PATCH/DELETE	/api/reminders	Read, mark, snooze, and delete reminders.
GET/POST/PUT/DELETE	/api/inventory	Manage stock records.
GET/PATCH	/api/alerts	Read alerts and mark as read.
GET/PATCH/DELETE	/api/notifications	Manage notifications.
GET	/api/reports/compliance	Get compliance and stock report data.

10. Data Flow Diagram



11. Data Design

The user-facing system works with user, profile, medicine, reminder, inventory, alert, notification, and report data. The frontend communicates with backend API endpoints through a centralized Axios integration layer.

Entity	Important Fields
User	fullName, email, mobile, token

Profile	age, gender, disease, emergency contacts, completion status
Medicine	medicineName, dosage, frequency, timings, startDate, notes
Reminder	medicineId, medicineName, dosage, date, time, status
Inventory	medicineName, currentStock, minimumStock, expiryDate, stockStatus
Alert	medicine, type, date/time, status, message
Notification	type, title, message, status, createdAt
Report	compliancePercentage, takenCount, missedCount, lowStockCount, outOfStockCount

12. Validation Rules

- Full name is required during registration.
- Mobile number must be a valid 10-digit number starting with 6, 7, 8, or 9.
- Email must follow valid email format.
- Password must contain at least 8 characters, one uppercase letter, one number, and one special character.
- Password cannot contain spaces and confirm password must match.
- Medicine and stock forms validate required fields before saving.

13. User Interface Design

- Healthcare-oriented green and blue visual identity.
- Dashboard provides summary cards, daily schedule, inventory overview, medicine list, and calendar.
- Sidebar navigation opens Home, Profile, Medicines, Reminders, Alerts, Reports, Inventory, and Logout.
- Modal forms are used for adding medicines, adding stock, completing profile, and confirming logout.
- Toast messages provide success and error feedback.
- Responsive layout supports desktop and mobile screens.

14. Testing Scope

- Authentication: registration validation, login, forgot password, reset password, logout, and session clearing.
- Profile: required fields, contact validation, save, and update.
- Medicines: add, edit, search, notes, frequency, timing, pagination, and delete.
- Reminders: display window, taken, snooze, history, and delete.
- Inventory: add stock, edit stock, delete stock, sorting, search, status chart, and low-stock calculation.
- Alerts and notifications: load, filter, search, mark read, dismiss, and clear.
- Reports: compliance percentage, taken/missed count, low-stock count, and out-of-stock count.

15. Limitations

- Actual email verification and password reset require backend and email service integration.
- Real-time alerts depend on backend availability and refresh behavior.
- The system supports tracking and reminders only; it does not provide medical advice.

16. Future Enhancements

- Add real-time push notifications for medicine reminders.
- Add downloadable reports in PDF or CSV format.
- Add wearable device integration for vitals tracking.
- Add caregiver sharing with explicit user consent.
- Add refill prediction based on stock and daily dosage.
- Add multilingual support for wider accessibility.

17. Conclusion

Healthcare Monitoring System provides a focused user portal for medicine management, reminders, inventory tracking, alerts, notifications, and compliance reporting. With User and System as the only documented stakeholders, the project remains centered on personal healthcare tracking and operational system support.